

Report with potential improvements Task 2

1. **Domain-specific fine-tuning:** Fine-tuning the feature extractor on Sentinel-2 multi-spectral bands (including Near-Infrared / NIR) instead of generic RGB to improve matching accuracy in agricultural and forested regions.
2. **Pipeline upgrades:** Evaluating LightGlue (SuperPoint + LightGlue) as an alternative matching backbone to reduce inference computation time while boosting feature precision.
3. **Preprocessing & quality filtering:** Integrating cloud and shadow masking using Sentinel-2 Scene Classification Layer (SCL) and applying CLAHE normalization to mitigate extreme seasonal light and snow variations.
4. **Geographical metric evaluation:** Transitioning reprojection error metrics from pixel-space distance to real-world geographic coordinates (meters) using GeoTIFF metadata.